CENTRE FOR QUANTUM TECHNOLOGIES
SINGAPORE

S14 Middleware

Group meeting

Elis Ortiz

Research Fellow — Quantum Middleware

CQT, National University of Singapore

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Measure → decide → continue

Mid-circuit results choose gates separately on every shot.

Vocabulary

- **Measure now**
  M(..., collapse=True) returns a result
- **Ask a condition**
  .is_one() . equals("10")
- **Choose a branch**
  IfElse(condition, then_do, else_do)
- **Match many values**
  Switch(result).case(...).default(...)

If measurement is 1 → apply X → final state is 0

```
from qibo import Circuit, gates

c = Circuit(1)
c.add(gates.H(0))

m = c.add(gates.M(0, collapse=True))
c.add( gates.IfElse(
        m.is_one(),
        then_do=gates.X(0),
    )
)

c.add(gates.M(0))
```

Three demonstrations

Each circuit measures mid-run, branches on that result, and continues.

01

Fast reset

1 qubit · IfElse



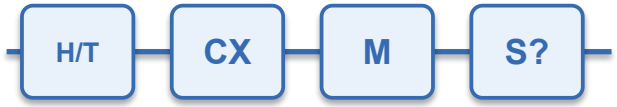
H → M → if 1: X → M

Active feedback on one qubit

02

T-gate teleportation

2 qubits · IfElse



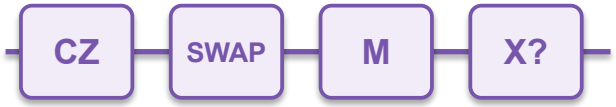
entangle → M → if 1: S

Correction from a measured bit

03

Repetition code

5 qubits · Switch



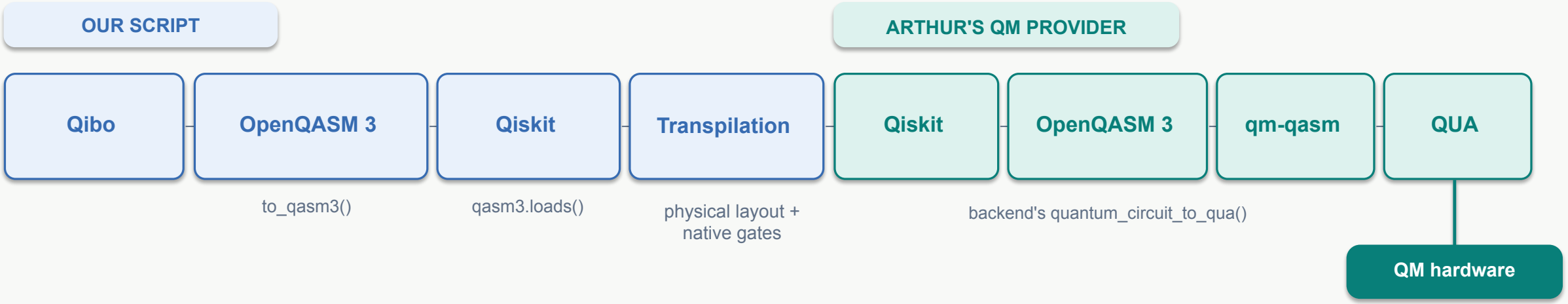
parity M → Switch → X

Two-bit syndrome selects a fix

Artifacts Simulation examples • Hardware demo notebook • GitHub Pages getting-started guide

From Qibo circuit to QUA program

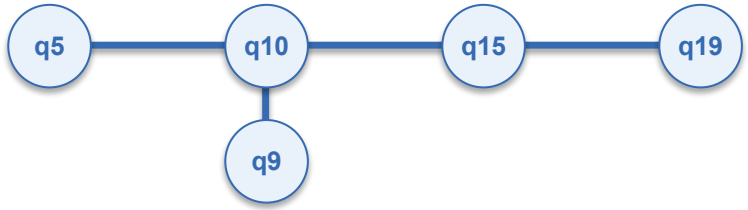
The Qiskit stage places qubits and translates gates to the QM target.



Explicit physical placement for the 5-qubit repetition circuit

Logical wires 0–4 → q5, q10, q9, q15, q19

Only these CZ edges are used; a q10–q9 SWAP moves states between parity checks.



Example: Repetition code on hardware

Injected X error on the middle data state · 1,000 shots · Slurm job 65369

635

of 1,000 shots

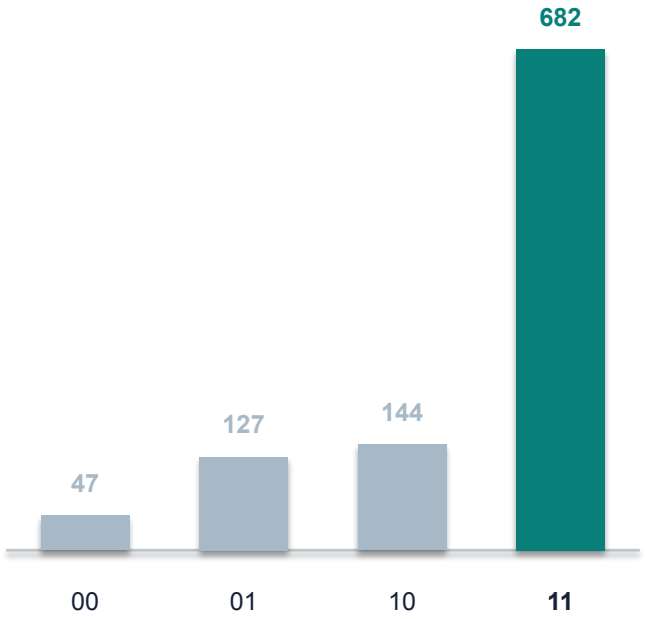
finished as 000

63.5% observed final-000 fraction

Single injected-error experiment; this could improve if we had a longer qubit chain and we could remove the SWAP

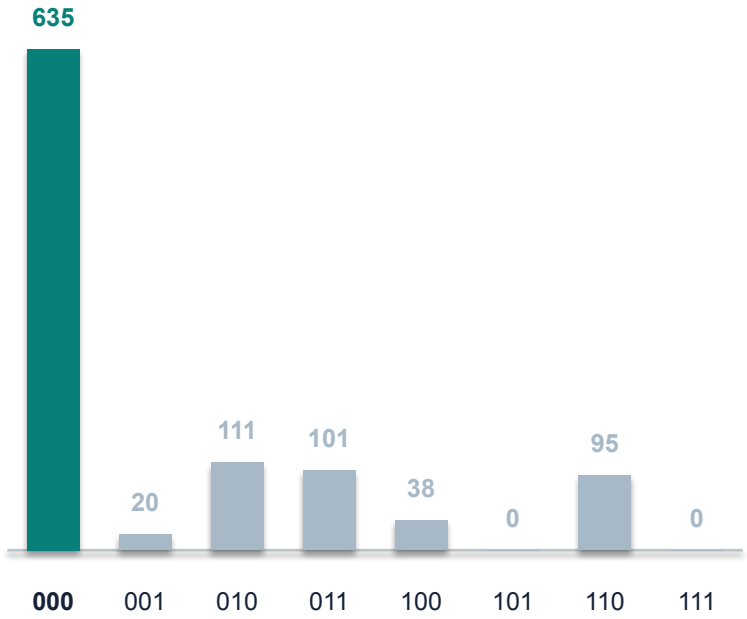
Syndrome

ancillas: q9, q15



Final data

data: q5, q10, q19



Expected signal: 11 syndrome → correction → 000 final data (observed together 602 times)